MILITARY SPECIFICATION

BALLISTICS COMPUTER SUBASSEMBLY: 10552928 (CANT UNIT)

1. SCOPE

1.1 This specification covers one type of electrical mechanical assembly known as Ballistics Computer Subassembly: 10552928 hereinafter referenced as "cant unit" (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of issue in effect on date of invitatior for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts Equipment and Tools for Ordnance Materiel: Packaging of
MIL-I-45607	Inspection Equipment Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U.S. Army, Frankford Arsenal

F10552928	Ballistics Computer Subassembly: 10552928 (Cant Unit)

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PACKAGING DATA SHEET

10552928 Packaging of Ballistics Computer Sub-assembly: (Cant Unit)

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials should be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.2 Fabrication.- The cant unit shall be manufactured in accordance with Drawing F10552928 and drawings pertaining thereto.

3.3 General specification.- The contractor shall be responsible for adherence to and compliance with the following requirements of MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.4 Environmental.

3.4.1 Storage temperature.- The cant unit shall show no evidence of physical failure and shall meet all requirements of this specification at standard ambient temperature of +60° to 90°F after having been exposed to and thermally stabilized at ambient temperature of -80° and +160°F.

3.4.2 Operating temperatures.- The cant unit shall meet all the requirements of 3.6 while exposed and thermally stabilized at ambient temperatures of -40° and +150°F. Upon return to standard ambient temperature the cant unit shall meet the requirements of 3.4.3.

3.4.3 Vibration.- The cant unit shall meet the requirements of 3.5 to 3.6.3 inclusive after being vibrated in a vertical plane for five minutes plus or minus 15 seconds at an amplitude of 1/16 inch (1/8 inch total excursion) at a frequency of 30 cycles per second.

3.4.4 Sealing.- When pressurized to an internal pressure of 5 ± 0.1 psig, the cant unit shall not exhibit a pressure drop greater than 0.2 psig for a period of at least 2 minutes.

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3.5 Electrical.

3.5.1 Continuity.- The continuity of the cant unit shall conform to the electrical schematic Drawing C10516514.

3.5.2 Dielectric strength.- There shall be no puncture of insulation, no sparkover, or decrease in dielectric strength when the insulated areas of the cant unit are subjected to an alternating current (ac) root mean square (rms) potential of 300 volts at a frequency of 60 cycles per second (cps) for a period of not more than 30 seconds nor less than five seconds.

3.5.3 Insulation resistance.- All insulation between ungrounded circuits and ground and between isolated circuits shall have a minimum resistance of 10 megohms with 500 volts dc applied.

3.6 Performance.

3.6.1 Resolver input travel.- The magnet arm shall contact the stops limiting travel, at cant unit housing rotations of 15 degrees minus zero plus 3.0 minutes, clockwise and counterclockwise, from a base level reference.

3.6.2 Magnet arm damping.- The magnet arm shall return to its normal gravity controlled position within a maximum of one cycle of oscillation when displaced at least 5 degrees.

3.6.3 Output voltages.- The cant unit output voltage ratios shall conform to the requirements of Table I in Drawing 10552928 when the unit is energized in conformance with the test circuit shown thereon. The minimum null indication, neglecting test circuit switching spikes, shall not exceed 8.0 millivolts peak-to-peak for cosine readings and 2.0 millivolts peak-to-peak for sine readings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

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4.2 First article (initial production) inspection.- When required by the contract, an initial production sample shall be submitted to a Government approved facility in accordance with the contract schedule for test and evaluation. The sample shall consist of three cant units plus three of every component and subassembly covered by Supplementary Quality Assurance Provisions (SQAP). The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures, as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to all requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials. After inspection by the contractor and provisional acceptance at source by the Government, the sample shall be inspected at a Government facility or such other facility as specified in the contract, for all the requirements of the drawings and specifications.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination, and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II, and III herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II, and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I CLASSIFICATION OF DEFECTS

CLASS	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 1.0% Defective		
101. Vibration	3.4.3	4.6.3
102. Sealing	3.4.4	4.6.4
103. Continuity	3.5.1	4.6.5
104. Dielectric strength	3.5.2	4.6.6
105. Insulation resistance	3.5.3	4.6.7

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TABLE I CLASSIFICATION OF DEFECTS (CONTINUED)

		REQUIREMENT	TEST PROCEDURE
106.	Resolver input travel	3.6.1	4.6.8.1
107.	Magnet arm damping	3.6.2	4.6.8.2
108.	Output voltages	3.6.3	4.6.8.3

MINOR: NONE

NOTE: The tests of Table I shall be conducted at a temperature between 60° and 90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One cant unit shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table II.

TABLE II

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.2	Applicable Drawing - Visual
302.	General specification	3.3	MIL-F-13926 - Visual

4.4.2 Environmental.- Three cant units shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table III, and shall then meet the requirements and tests in Table I.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Storage temperature (-80° and +160°F)	3.4.1	4.6.1
304.	Resolver input travel	3.4.2, 3.6.1	4.6.2, 4.6.8.1
305.	Magnet arm damping	3.4.2, 3.6.2	4.6.2, 4.6.8.2
306.	Output voltages	3.4.2, 3.6.3	4.6.2, 4.6.8.3

NOTE: The tests for Characteristics 304 through 306 inclusive shall be conducted at -40 and +150°F.

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4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Sealing.- The pressure test equipment to test sealing shall be capable of internal pressurizing the cant unit, with clean, dry, air, to the psig value specified. The pressure gauge of the test equipment shall be graduated to measure any variation of the internal pressure in increments of 0.1 psig.

4.5.3 Test equipment.- In conjunction with 4.5.2.2 the following standard test equipment shall be utilized in the performance of the applicable tests as specified in 4.6.

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ITEM NUMBER		DESCRIPTION
1.	Ratio standard	Ratio range from .00000 to 1.00000 accuracy to ± .01%. Input impedance at 400 cps, 10,000 ohms minimum. Series impedance - 10 ohms maximum. Phase shift 0.01° maximum.
2.	Oscilloscope	Minimum sensitivity of 20 millivolts, full scale, vertical channel. External synchronization. Input impedance - 1 megohm minimum.
3.	A.C. power supply	0-45 volts rms, 400 cps ± 2 cps, 0.5% maximum harmonic distortion, maximum variation in distortion during test 0.05%.
4.	Testing fixture	Shall mount the cant unit in a true horizontal position. Rotate the cant unit up to 20° in either direction about the resolver axis with settings to 5°, 10° and 15° within 20 seconds. The test fixture shall also contain a setting of 15 degrees 3 minutes. Provision for instantaneous release from a 10° position and return to horizontal position.

4.6 Test methods and procedures.

4.6.1 Storage temperature.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.1. The cant unit shall be exposed to each temperature extreme specified in 3.4.1 for a minimum of 5 hours. After each temperature exposure the cant unit shall be stabilized in an ambient temperature between 60° and 90°F and subjected to the visual, tactile, and audible examination of 4.6.3.1. The cant unit shall then be subjected to the tests in 4.6.2 through 4.6.8.3 inclusive to determine conformance to 3.4.1. Caution shall be exercised during the temperature cycling to avoid subjecting the cant unit to thermal shock.

4.6.2 Extreme operating temperature.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.2. The cant unit shall be exposed to each temperature extreme specified in 3.4.2 for a period of 5 hours. At the end of each 5 hour period, and while still at the temperature extreme, the cant unit shall be subjected to the applicable test of 4.6.8.1 through 4.6.8.3 inclusive. The cant unit shall conform to the requirements of 3.4.2. Caution shall be exercised during this test to avoid subjecting the cant unit to thermal shock. The test circuitry utilized to energize the cant unit shall be positioned external to the environmental test chamber. Upon completion of this test the cant unit shall be subjected

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to the test in 4.6.3.

4.6.3 Vibration.- Standard vibrating equipment capable of providing the amplitude and frequency specified in 3.4.3 shall be utilized for the vibration test. Unless otherwise specified the test equipment shall be in conformance with the "Test Facilities" requirements of MIL-F-13926. The cant unit shall be mounted on the vibrating table by means of an adapter. This adapter shall position the cant unit with its mounting surfaces in a horizontal plane. The cant unit shall be vibrated in a vertical plane for a period of 5 minutes ± 15 seconds. Upon completion of the vibration cycle the cant unit shall be subjected to the tests specified in 4.6.3.1.

4.6.3.1 Examination.- Subsequent to vibration the cant unit shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine conformance with 3.4.3, prior to being subjected to the remaining tests specified herein. A visual examination shall be made for missing or damaged parts. A tactile examination shall also be conducted to determine looseness of parts such as the connector and covers. An audible examination shall be made of the cant unit by shaking it and listening for loose internal parts or broken components, such as screws, stops, magnets, etc. Any one of the possible defects defined above shall be cause to consider a cant unit defective. Cant units which have passed the above examinations shall be subjected to the tests specified in 4.6.5 through 4.6.8.3 inclusive to determine overall compliance with 3.4.3.

4.6.4 Sealing.- The internal pressurizing of the cant unit to test sealing shall be performed with testing equipment conforming to 4.5.2.3.1. For this test remove the Plug MS-49005-1 from the cant unit and replace it with an air valve containing a spring loaded valve stem. The pressure shall be applied to the cant unit through this valve. Pressurize the cant unit with clean, dry, air to the psig value specified in 3.4.4. The applied pressure shall be retained for a minimum of 2 minutes. Any evidence of leakage in the cant unit at the end of the 2 minute time period shall not exceed the tolerance specified in 3.4.4, as measured on the pressure gauge of the testing equipment. Upon completion of this test remove the air valve and replace the Plug MS-49005-1 in the cant unit.

4.6.5 Continuity.- A continuity test of the cant unit circuits shall be made using standard measuring equipment to determine conformance with the requirements of 3.5.1.

4.6.6 Dielectric strength.- The cant unit shall be tested in accordance with Method 301 of MIL-STD-202 and shall meet the requirements specified in 3.5.2.

4.6.7 Insulation resistance.- Immediately after satisfactorily withstanding the test for dielectric strength, the cant unit shall be given the insulation resistance test in accordance with Method 302,

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test condition B, of MIL-STD-202, and snall meet the requirements specified in 3.5.3.

4.6.8 Performance.- These tests shall be performed using the applicable testing equipment specified in 4.5.3, and in accordance with the test circuit of Drawing F10552928.

4.6.8.1 Resolver input travel.- To determine conformance to 3.6.1, physically rotate the cant housing in a clockwise direction about the centerline of the resolver axis, by means of the test fixture. Note and record the angle on the test fixture at which the output voltage of the cant unit becomes a constant value. Return the fixture to zero cant, and repeat the above test by rotating the cant housing in a counterclockwise direction while observing its output voltage.

4.6.8.2 Magnet arm damping.- To determine conformance to 3.6.2, physically rotate the cant housing 10 degrees about the centerline of the resolver axis. At this point, and instantaneous release of the test fixture and return of the cant housing to the original horizontal position of the locating surfaces shall cause the magnet arm to return to its normal gravity controlled position within one cycle of oscillation as specified in 3.6.2. One cycle of oscillation shall be determined by observing the oscilloscope null on the test circuit.

4.6.8.3 Output voltages.- To determine conformance to 3.6.3 mount the cant unit to the testing device specified in 4.4.2 and connect the unit to the test circuit specified in 4.6.8. With the proper input voltage and ratio standard setting, check the output on the oscilloscope for square wave display. Set the cant unit to 0 degrees and position S1 and S2 switches as shown on Table I of Drawing F10552928. Observe ratio setting for conformance to the tolerance specified. Continue to test for all right and left cant angles listed in the referenced Table I.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10552928. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The cant unit is primarily intended as a functional component of the electronic Computer, Ballistics: M16. The cant unit is designed to measure gun trunnion cant angles, convert these angles into electrical voltage functions of sine and cosine and supply the functions to the computer system where they are used to multiply ballistics correction factors when the weapon is sighted on target with the gun trunnion canted.

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6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Level of protection
(c) Applicable packaging data sheets (see 5.1)
(d) Applicable stock number

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1220-A068

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